

NAME: _____ MATRIC
NUMBER: _____

SCHOOL OF MATERIALS SCIENCE & ENGINEERING

CONTINUOUS ASSESSMENT 2 SEMESTER 2 2023-2024

MS1016 – THERMODYNAMICS OF MATERIAL

11 April 2023

Time Allowed: 1 hour

INSTRUCTIONS

1. This paper contains **THREE (3)** short questions and **TEN (10)** MCQ questions.
 2. Answer **ALL** questions.
 3. This is an open book test.
 4. All questions have **ONLY ONE** correct answer for the MCQ questions.
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PART I – SHORT QUESTIONS

- (1) Derive the Clausius-Clapeyron equation for the solid and vapour phase in equilibrium?

(20 marks)

Clapeyron Equation – for single component system

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{\Delta H}{T\Delta V}$$

For a solid transforming into vapour, the volume of the material increases many times as $V_{\text{vapour}} \gg V_{\text{solid}}$. $\therefore \Delta V$ is approximately V_{vapour} . Clapeyron approximates,

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{\Delta H}{TV}$$

Assuming the vapour approximates an ideal gas and given $PV = RT$, $V = RT/P$

$$\left(\frac{dP}{dT}\right)_{eq} = \frac{P\Delta H}{RT^2} \quad \text{or} \quad \left(\frac{dP}{P}\right)_{eq} = \left(\frac{\Delta H}{R}\right) \frac{dT}{T^2}$$

$$\text{Integrating } \ln[P]_{P_1}^{P_2} = -\left(\frac{\Delta H}{R}\right) \left[\frac{1}{T_2} - \frac{1}{T_1}\right]$$

$$\frac{P_2}{P_1} = \left(\frac{\Delta H}{R}\right) \left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

- =
2. Described the use and difference between the Raoult's Law, Henry's Law and the Gibbs-Duhem equation in solid solution.

(20 marks)

The activity coefficient, γ is defined as the ration of the activity over mole fraction,

$$\gamma = \frac{a}{X}$$

where a is the activity and X , the mole fraction. The **significance of Raoult's and Henry's law** is that calculation of the chemical potential of a component in a dilute solid solution can be simplified to the involvement of only the mole fraction. Given the chemical potential of a component, μ is

$$\mu = \underline{G} + RT \ln a$$

Raoult's law states that the activity of the major component in a dilute solution is approximately ideal and may be approximated to the mole fraction. Hence the chemical potential may be written as:

$$\mu = \underline{G} + RT \ln X$$

For the minor component in the dilute solution, the activity equals to γX . Henry's law states that γ may be approximated to be a constant because the composition is X is low. Hence

$$\mu = \underline{G} + RT \ln \gamma X$$

Therefore. Using Raoult's law and Henry's law, the chemical potentials of the components in a dilute solution may be calculated without using the activity.

The **Gibbs-Duhem equation** is use for calculation of the activities of the components in non-dilute solution.

$$d(\ln a_A) = -\frac{X_B}{X_A} d(\ln a_B)$$

For a binary solution, if the activity of one component is known, the other can be calculated using the Gibbs-Duhem equation.

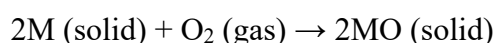
3. Consider the Ellingham diagram. The slope is positive for the majority of metal oxidation in the Ellingham diagram. Explain why and what will happen to these oxides when they are heated up to very high temperatures?

(20 marks)

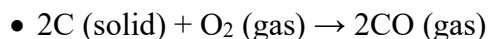
The positive slope for most metals means that the standard Gibbs free energy, ΔG^0 , of the oxidation of these metals increases with temperature.

$$\Delta G = \Delta H - T\Delta S$$

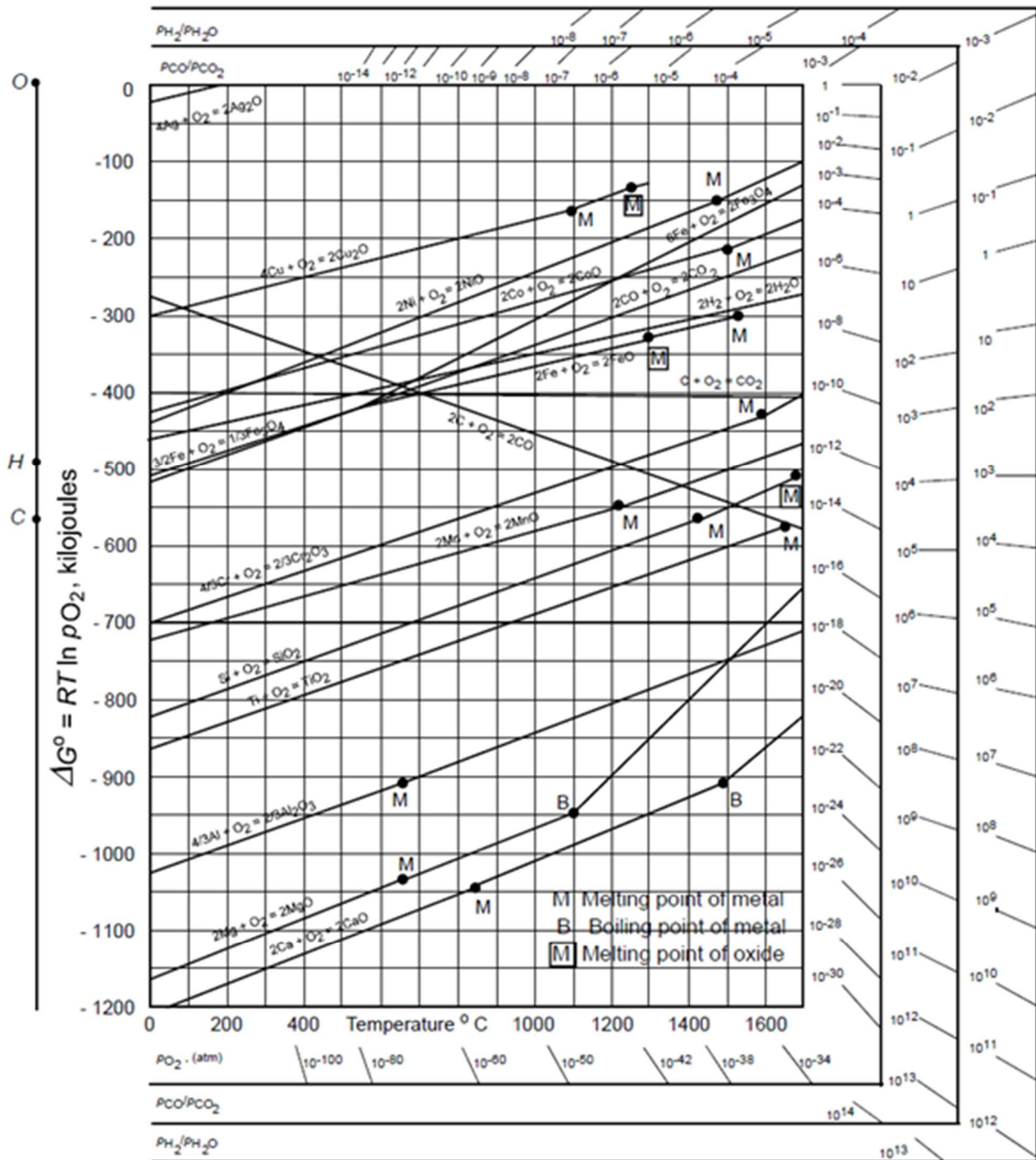
Consider the oxidation of a metal, M, oxidized by oxygen, O₂,



- The formation of metal oxides consumes oxygen gas, resulting in a decrease in entropy, meaning, ΔS becomes negative, and the term $T\Delta S$ in the straight-line equation becomes positive.
- Carbon monoxide formation is shown by a straight line with a negative slope. In this scenario, ΔS is positive because the consumption of one mole of oxygen gas results in two moles of CO gas. It implies that CO becomes more stable at higher temperatures.



- As the temperature rises, the ΔG value for the formation of metal oxide becomes less negative until it reaches zero at a certain point. Below this temperature, ΔG is negative and the oxide is stable; above this temperature, ΔG is positive and the oxide is unstable. This overall pattern shows that as temperatures rise, metal oxides become less stable and decompose more easily.



PART II MCQ QUESTIONS

- (4) Answer all the **TEN MCQ** questions below. There is only **ONE** answer for each question.

(40 marks)

- (a) Estimate the maximum temperature in which Mg can be used to reduce Al_2O_3 ?

- A 1600°C
- B 1500°C
- C 1400°C
- D 1300°C
- E 1200°C

ANS: B

- (b) Which of the following thermodynamics will not directly affects the equilibrium condition in a single component system?

- A molar change in volume
- B molar enthalpy of transformation
- C pressure
- D temperature
- E Gibbs free energy

ANS: E

- (c) Consider the reaction:

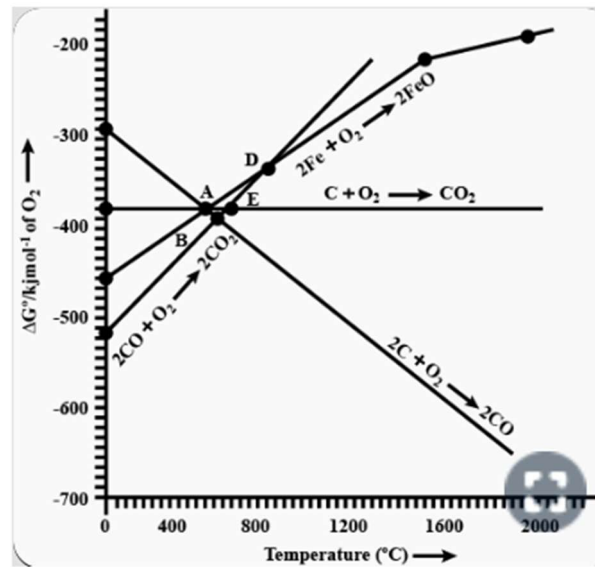


Which of the following statements is true?

- A Fe will spontaneously bond with C
- B Fe and C will not bond under any conditions
- C increasing temperature will promote the reaction
- D increasing temperature will impede the reaction
- E decreasing the temperature will result in the formation of Fe_3C spontaneously

ANS: C

- (d) Refer to the figure below and select the correct choice. At temperature below point A, FeO can _____



- A be reduced by CO and CO₂
- B be reduced by carbon only
- C not be reduced by C and CO
- D be reduced by CO only
- E not be reduced at all
- (e) Which of the following is the conditions for the formation of intermetallic?
- A $\Delta S_{\text{mix}} > 0$ and $\Delta H_{\text{mix}} > 0$
- B $\Delta S_{\text{mix}} < 0$ and $\Delta H_{\text{mix}} > 0$
- C $\Delta S_{\text{mix}} < 0$ and $\Delta H_{\text{mix}} < 0$
- D Low temperature and $\Delta H_{\text{mix}} > 0$
- E Low temperature and $\Delta H_{\text{mix}} < 0$

ANS: D

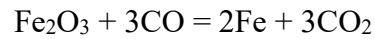
ANS: E

(f) The activity of a gaseous phase is approximately equal to the _____ of the gas.

- A partial pressure
- B partial volume
- C composition
- D temperature
- E enthalpy

ANS: A

(g) The equilibrium constant for the following reaction is:



A $\frac{a_{\text{Fe}}^2 \cdot P_{\text{CO}_2}^3}{a_{\text{Fe}_2\text{O}_3} \cdot 3P_{\text{CO}}}$

B $\frac{a_{\text{Fe}_2\text{O}_3}^2 \cdot P_{\text{CO}_2}^3}{a_{\text{Fe}_2\text{O}_3} \cdot P_{\text{CO}}}$

C $\frac{a_{\text{Fe}_2\text{O}_3} \cdot a_{\text{CO}}^3}{a_{\text{Fe}}^2 \cdot P_{\text{CO}_2}^3}$

D $\frac{a_{\text{Fe}}^2 \cdot P_{\text{CO}_2}^3}{a_{\text{Fe}_2\text{O}_3} \cdot P_{\text{CO}_2}}$

E $\frac{a_{\text{Fe}}^2 \cdot P_{\text{CO}_2}^3}{a_{\text{Fe}_2\text{O}_3} \cdot P_{\text{CO}}^3}$

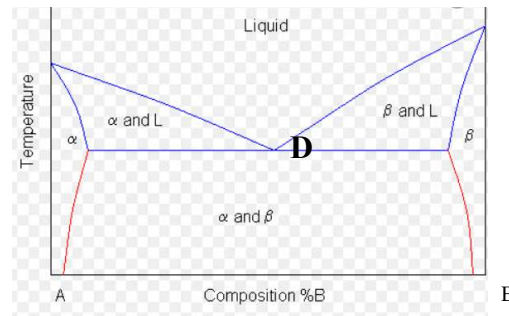
ANS: E

(h) Which of the following will not normally be regarded as a phase in an alloy containing components A and B?

- A 0.1% A dissolved in B
- B inclusions
- C 0.1% B dissolved in A
- D compounds of A and B
- E 50%A + 50%B

ANS: B

- (i) Which of the following is TRUE of the equilibrium phase diagram below?



- A $\Delta H_{\text{mix}} < 0$
- B $\Delta S_{\text{mix}} < 0$
- C 3 phases are in equilibrium at point D
- D melting point of A $>$ B
- E α phase is a dilute solutions of A in B
- (j) Which of the following statements best describe the chemical potential of a species in a mixture?
- A the energy that can be absorbed or released due to a change in temperature
- B the energy that can be absorbed or released due to a change in pressure
- C the energy that can be absorbed or released due to a change volume
- D the energy that can be absorbed or released due to a change of the particle number of the given species
- E change in molar Gibbs free energy of all the components

ANS: C

ANS: D